English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability and Statistics, Computer Science
Test 1, 2021/2022: solutions

English transcription of the supplied handwritten solution notes. Only the workings present in the notes are reproduced.

Problem 1

Let I denote the ordinary six-sided die. For an announced value $k \le 6$,

$$\mathbb{P}(I \mid \text{announced } k) = \frac{\frac{1}{6}\cdot\frac{1}{2}}{\frac{1}{6}\cdot\frac{1}{2}+\frac{1}{2}\cdot\frac{1}{8}} = \frac{1}{1+3/4} = \frac{4}{7}.$$

The probability is zero otherwise.

Problem 2

The attending students are represented by the counts

$$\text{Year 1: } \frac{4}{5}\cdot 600 = 480, \qquad \text{Year 2: } \frac{3}{5}\cdot 400 = 240, \qquad \text{Year 3: } \frac{1}{4}\cdot 200 = 50.$$

The first calculation in the notes uses

$$\frac{50(480+240)}{480\cdot 240 + 480\cdot 50 + 240\cdot 50}.$$

A more formal approach conditions on attendance:

$$\mathbb{P}(\text{Year 1} \mid \text{attends}) = \frac{\frac{4}{5}\frac{600}{1200}}{\frac{4}{5}\frac{600}{1200}+\frac{3}{5}\frac{400}{1200}+\frac{1}{4}\frac{200}{1200}} = \frac{48}{77},$$

$$\mathbb{P}(\text{Year 2} \mid \text{attends}) = \frac{24}{77}, \qquad \mathbb{P}(\text{Year 3} \mid \text{attends}) = \frac{5}{77}.$$

The desired conditional probability is that the higher-year student is in Year 3, given attendance and different years. The notes combine the Year 3 / Year 1 and Year 3 / Year 2 cases in the numerator, and add the Year 1 / Year 2 case in the denominator.

Transcription note: the handwritten final fraction is 15/103*. It does not agree with the displayed counts and conditional probabilities above; this discrepancy is already present in the source.*

Problem 3

The handwritten notes give

$$Y \sim Bin(10, 1/4), \qquad X = 10 + Y.$$

Problem 4

(a) Direct transmission

$$\mathbb{P}(\text{no errors}) = (1-p)^8, \qquad \mathbb{E}[Bin(8, 1-p)] = 8(1-p).$$

(b) Three intermediate servers

The message travels through four transmissions, so a bit is received correctly after zero, two or four errors. Thus

$$\mathbb{P}(\text{bit } i \text{ is correct}) = (1-p)^4 + \binom{4}{2}(1-p)^2p^2 + p^4 =: q.$$

Consequently,

$$\mathbb{P}(\text{the message is correct}) = q^8, \qquad \mathbb{E}[Bin(8,q)] = 8q.$$

For $2n$ transmissions,

$$q = \sum_{k=0}^{n} \binom{2n}{2k} p^{2k}(1-p)^{2n-2k}.$$

Using

$$\sum_{k=0}^{2n} \binom{2n}{k} p^{k}(1-p)^{2n-k} = 1$$

and

$$\sum_{k=0}^{2n} \binom{2n}{k} p^{k}(1-p)^{2n-k}(-1)^k = (p-(1-p))^{2n},$$

the notes conclude

$$2q = 1 + (2p-1)^{2n}, \qquad q \longrightarrow \frac{1}{2} \quad \text{as } n \to \infty, \quad |2p-1| < 1.$$